

Milestone 2 Report

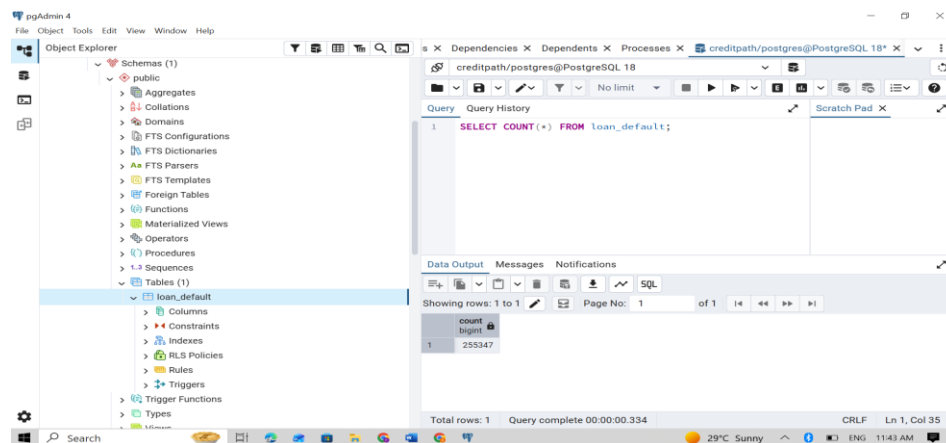
CreditPathAI – Data Ingestion & Exploratory Data Analysis (EDA)

1. Introduction

The objective of Milestone 2 is to perform data ingestion and exploratory data analysis (EDA) on the loan default dataset. The dataset contains financial and demographic information about borrowers which will later be used to build a machine learning model to predict loan defaults. In this stage, the raw CSV dataset was stored in a PostgreSQL database and analyzed using Python libraries such as Pandas, Matplotlib, and Seaborn.

2. Data Ingestion

The raw dataset was first imported into PostgreSQL to follow proper data engineering practices. A database named 'creditpath' was created and a table called 'loan_default' was used to store the dataset. The dataset was imported from CSV using pgAdmin's Import/Export tool.



Result: 255,347 rows successfully inserted into the database.

3. Data Loading into Python

After ingestion, the dataset was loaded into Python using SQLAlchemy and Pandas to perform analysis and visualization.

4. Dataset Overview

The dataset contains 255,347 records and 18 features describing borrower financial information such as age, income, loan amount, credit score, employment details, and loan purpose. The target variable is 'default_status', where 0 indicates no default and 1 indicates loan default.

5. Exploratory Data Analysis (EDA)

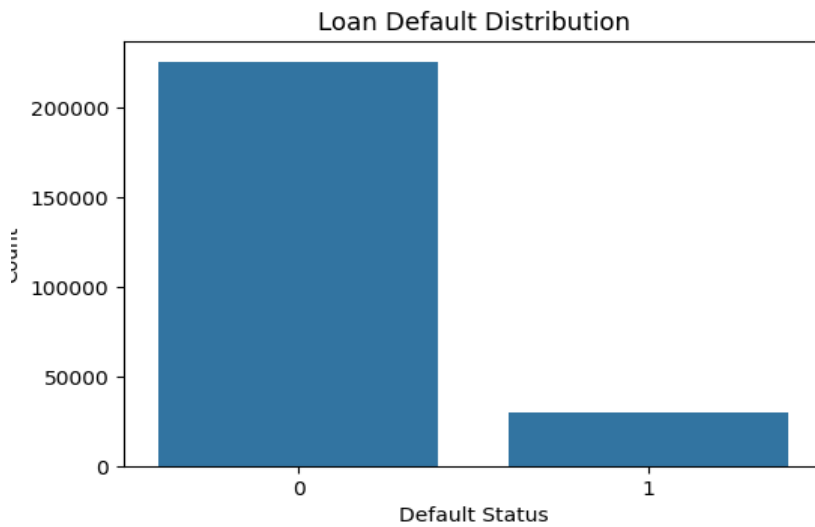
EDA was performed to understand the structure, patterns, and relationships in the dataset. Basic inspection methods such as `df.head()`, `df.info()`, and `df.describe()` were used to analyze data types,

distributions, and statistical summaries. Missing values were checked using `df.isnull().sum()` and none were found in the dataset.

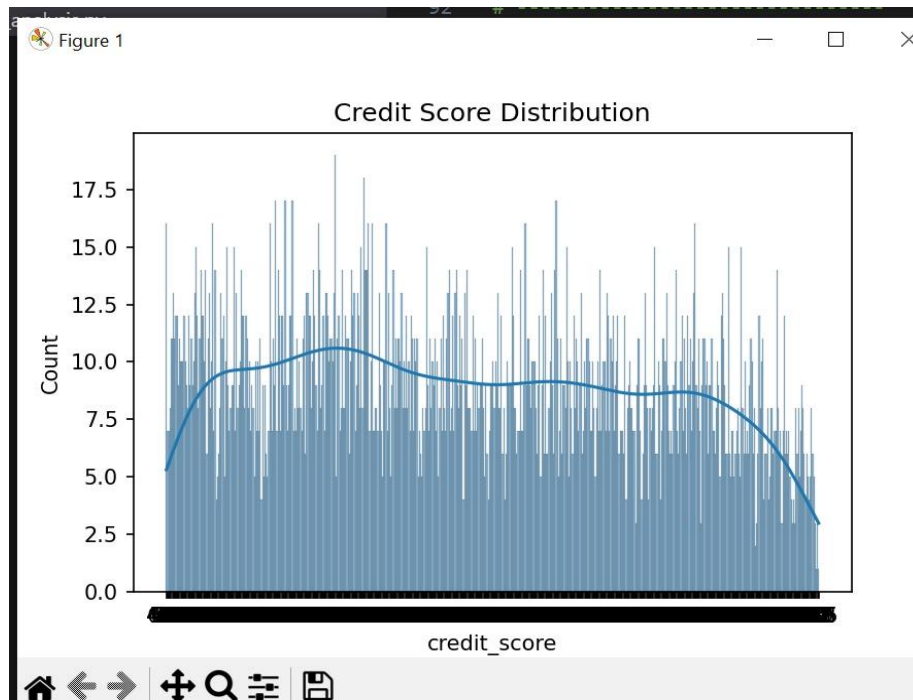
6. Visualizations

The following visualizations were generated during EDA:

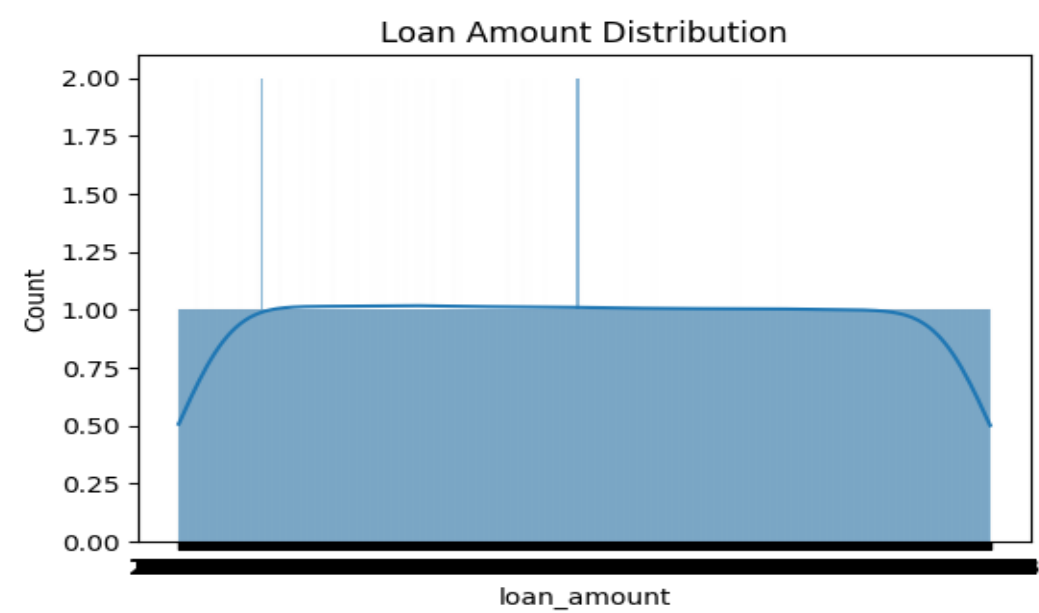
Loan Default Distribution



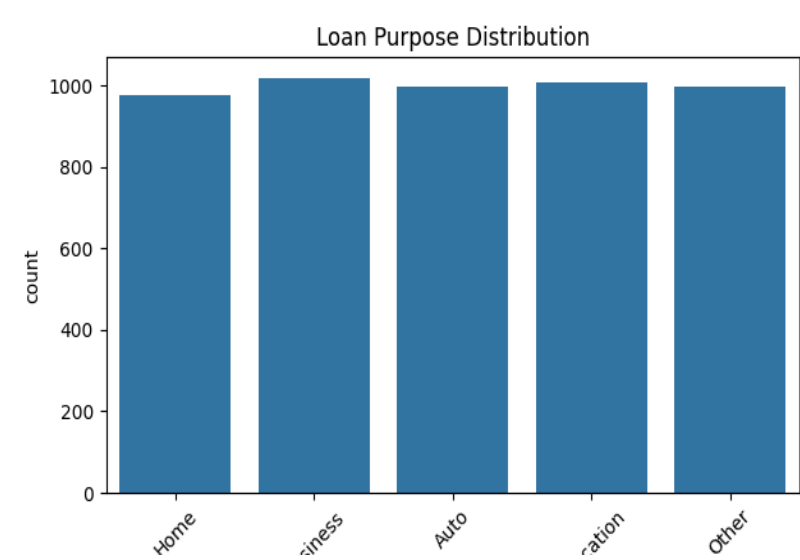
Credit Score Distribution



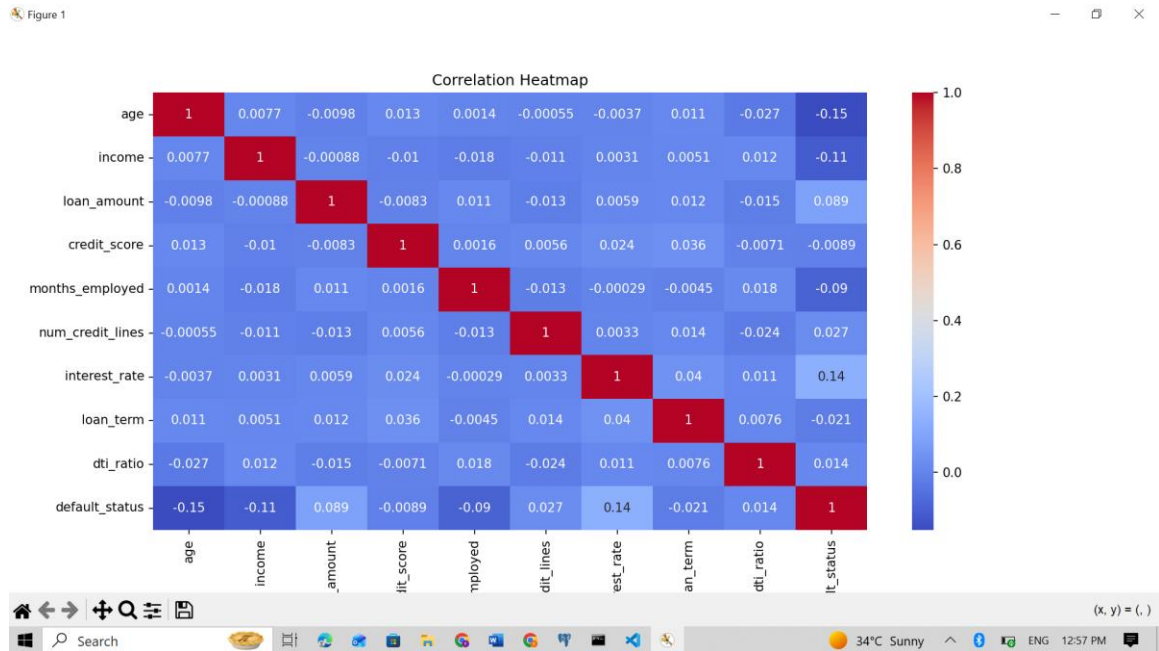
Loan Amount Distribution



Loan Purpose Distribution



Correlation Heatmap



7. Data Cleaning

To prepare the dataset for machine learning, preprocessing steps were performed. Duplicate records were checked and removed if present. Since no missing values were detected, no imputation was required. Categorical columns such as education, employment_type, marital_status and loan_purpose were converted into numeric format. The column 'loan_id' was removed because it is only an identifier and does not contribute to predicting loan default.

The cleaned dataset was saved as 'clean_loans.csv' for use in the next milestone which involves training machine learning models.

8. Conclusion

In Milestone 2, the loan dataset was successfully ingested into PostgreSQL and analyzed using Python. Exploratory data analysis provided insights into borrower financial patterns and loan default distribution. The dataset was cleaned and prepared for machine learning, making it ready for the next stage of the project which involves building predictive models.